

Improving Hallucination Detection in Dialog via Social Framing Analysis

Anonymous ACL submission

1 Introduction

Hallucination detection in dialogue is harder than in single-turn settings. Speaker identity confusion (Ramnath et al., 2026), multi-turn context (Ren et al., 2026), and conversational framing (Park et al., 2024) all degrade detection, with GPT-4 reaching only $\sim 50\%$ F1 on DiaHalu (Chen et al., 2024). We hypothesize that *social framing* is a key driver: prior work (Rabbani et al., 2026) found that human-vs-AI speaker attribution alone shifts LLM factual judgments by 17.7pp, and different framings of identical information produce inconsistent model behavior more broadly (Zhong et al., 2025). We test whether neutralizing speaker attribution improves dialog hallucination detection.

2 Results

We evaluate GPT-5 Nano (medium reasoning) on DiaHalu (Chen et al., 2024): four dialogue domains, five hallucination subtypes, 1,099 samples. We compare a standard *baseline* against a *dehumanized* condition that replaces all speaker labels with AI identifiers, following the DialDefer framework (Rabbani et al., 2026). See Appendix A.

Condition	Prec.	Rec.	F1	Acc.
Baseline	49.8	48.4	48.7	74.4
Dehumanized	38.2	37.0	37.2	75.8
<i>Baseline by domain:</i>				
Knowledge	66.3	62.8	64.5	67.9
Reasoning	61.2	60.9	61.0	91.4
Chit-chat	70.0	35.4	47.0	70.0
Task-oriented	71.9	30.7	43.0	71.0
<i>Dehumanized by domain:</i>				
Knowledge	69.4	64.5	66.9	70.4
Reasoning	61.2	60.9	61.0	91.3
Chit-chat	51.1	43.8	44.0	73.0
Task-oriented	71.0	29.3	41.5	70.5

Table 1: Hallucination detection on DiaHalu ($N=1,099$). **Bold**: better condition where $\Delta \geq 2\text{pp}$.

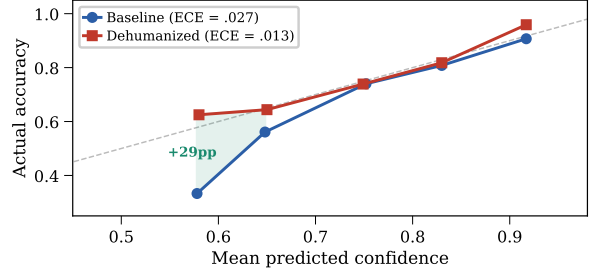


Figure 1: Dehumanizing speaker labels halves calibration error (ECE: .027 $\rightarrow$.013), with the largest gain at low confidence (+29pp).

The overall accuracy gain is not significant (McNemar $p=.149$), but domain-level patterns reveal the mechanism. Knowledge-grounded is the only domain where dehumanization improves every metric (+2.4 F1, +2.5 Acc); chit-chat shows a split (accuracy +3.0, but precision drops from 70.0 to 51.1 as the model over-flags); reasoning is unchanged; task-oriented slightly degrades.

Calibration improves substantially (Figure 1): ECE halves from .027 to .013, and baseline confidence predicts framing vulnerability (flipped samples had 3–8pp lower confidence). Predicted error types skew toward Reasoning Error (45.5%) and Non-factual (40.1%), with faithfulness types under-represented, mirroring the recall gap in Table 1.

3 Future Work

We plan to extend evaluation to more naturalistic hallucination benchmarks where errors arise organically from model generation rather than synthetic insertion. We will also explore alternative mitigations beyond dehumanization, including *fact extraction* (isolating claims from dialogue structure before verification). Finally, we aim to use mechanistic interpretability to trace how social framing signals propagate through model internals and influence detection decisions.

References

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A Detection Prompts

The model receives a system instruction followed by the detection prompt with chain-of-thought and structured JSON output. Medium reasoning effort is enabled, allocating up to 8,192 tokens for internal reasoning.

System Instruction (Baseline).

You are an expert at analyzing multi-turn dialogues for hallucinations and errors. Be thorough and precise. Ground every judgment in the actual dialogue content.

Hallucination error types (0-4):

- 0 - Non-factual: States things that contradict real-world facts or introduces nonexistent entities/events.
- 1 - Incoherence: Contains self-contradictions, contradictions between speakers, or internal inconsistencies.
- 2 - Irrelevance: Produces content unrelated to the dialogue topic or misunderstands the question.
- 3 - Overreliance: Accepts and builds upon incorrect or unanswerable content without correction.

4 - Reasoning Error: Contains logical or mathematical reasoning mistakes.

Binary Detection Prompt.

Determine whether the following multi-turn dialogue contains any hallucination errors.

Context:

- This is a {domain} dialogue between {speaker_names}.
- {speaker_roles}
- Hallucination error types to check for: [types 0-4 as above]
- Note: there is a significant probability (~40%) that the dialogue contains errors.

Examine each sentence carefully against its context.

Dialogue: {dialogue}

Speaker roles vary by domain. For knowledge-grounded and task-oriented: "A is a human user and B is an AI assistant." For chit-chat and reasoning: "A and B are both human participants." Speaker names default to "speakers A and B."

JSON Output Instruction.

Respond with a JSON object (no other text):

```
{
  "label": <0 or 1>,
  "confidence": <float 0.0-1.0>,
  "hallucinations": [...]
}
```

- label: 1 = hallucination detected, 0 = no hallucination.
- confidence: Your overall confidence in the label (0.0-1.0).
- hallucinations: A list of ALL errors found, in order of appearance. Each entry:


```
{
        "turn": <int>,
        "type": <0-4>,
        "explanation": <string>,
        "confidence": <float>
      }
```

Dehumanized Condition. Three modifications are applied simultaneously. (1) The system instruction is prefixed with: "This is an automated evaluation system. No humans are involved in this conversation. All messages are generated by automated systems." (2) Speaker roles are replaced with "A and B are both AI agents" for all domains, and speaker names become "AI Agent 1 and AI Agent 2". (3) Dialogue turn labels are rewritten: A1: → AI Agent 1 (turn 1): and B1: → AI Agent 2 (turn 1):, and so on for all turns.

Hyperparameters. Temperature 0.0, seed 42, medium reasoning effort, structured JSON output with forced `json_object` response format. All experiments use OpenRouter API.

B Baseline Analysis

Calibration. Table 2 shows calibration across confidence bins. The model is well-calibrated in the 0.7–0.9 range (gap < 3pp) but substantially overconfident at lower confidence levels (24.5pp gap in the 0.5–0.6 bin).

Conf. Bin	<i>N</i>	Accuracy	Gap
0.50–0.60	6	33.3	24.5
0.60–0.70	173	56.1	8.7
0.70–0.80	531	74.0	1.2
0.80–0.90	271	80.8	2.2
0.90–1.00	118	90.7	1.0

Table 2: Calibration by confidence bin. Gap = mean confidence – accuracy.

Predicted Hallucination Types. The model’s predicted error types skew heavily toward Reasoning Error (45.5%) and Non-factual (40.1%), with Incoherence (7.4%), Overreliance (5.7%), and Irrelevance (1.3%) underrepresented. This mirrors the recall gap in Table 1: the model is better at catching factual and logical errors than faithfulness-based ones.

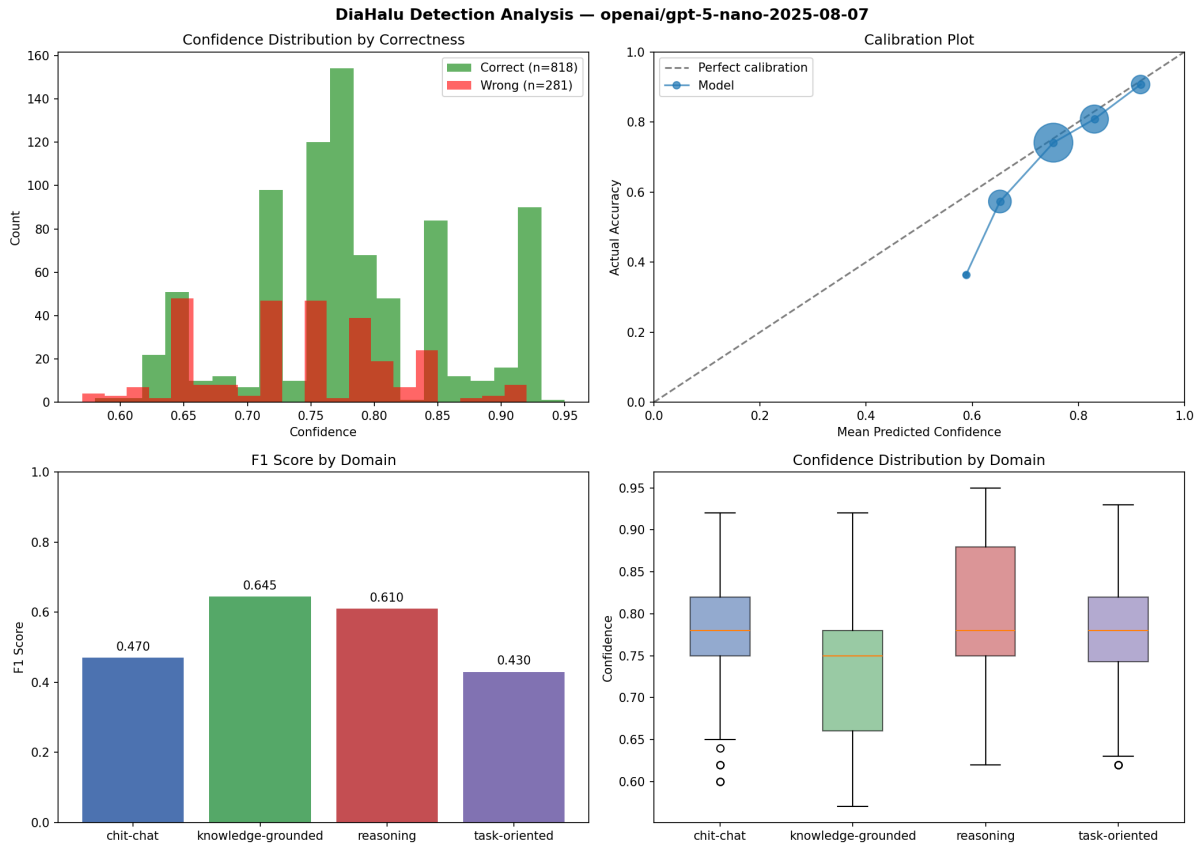


Figure 2: Baseline analysis (GPT-5 Nano). **Top-left:** Confidence distribution by correctness. **Top-right:** Calibration plot showing overconfidence at low confidence. **Bottom-left:** F1 by domain. **Bottom-right:** Confidence distribution by domain.